

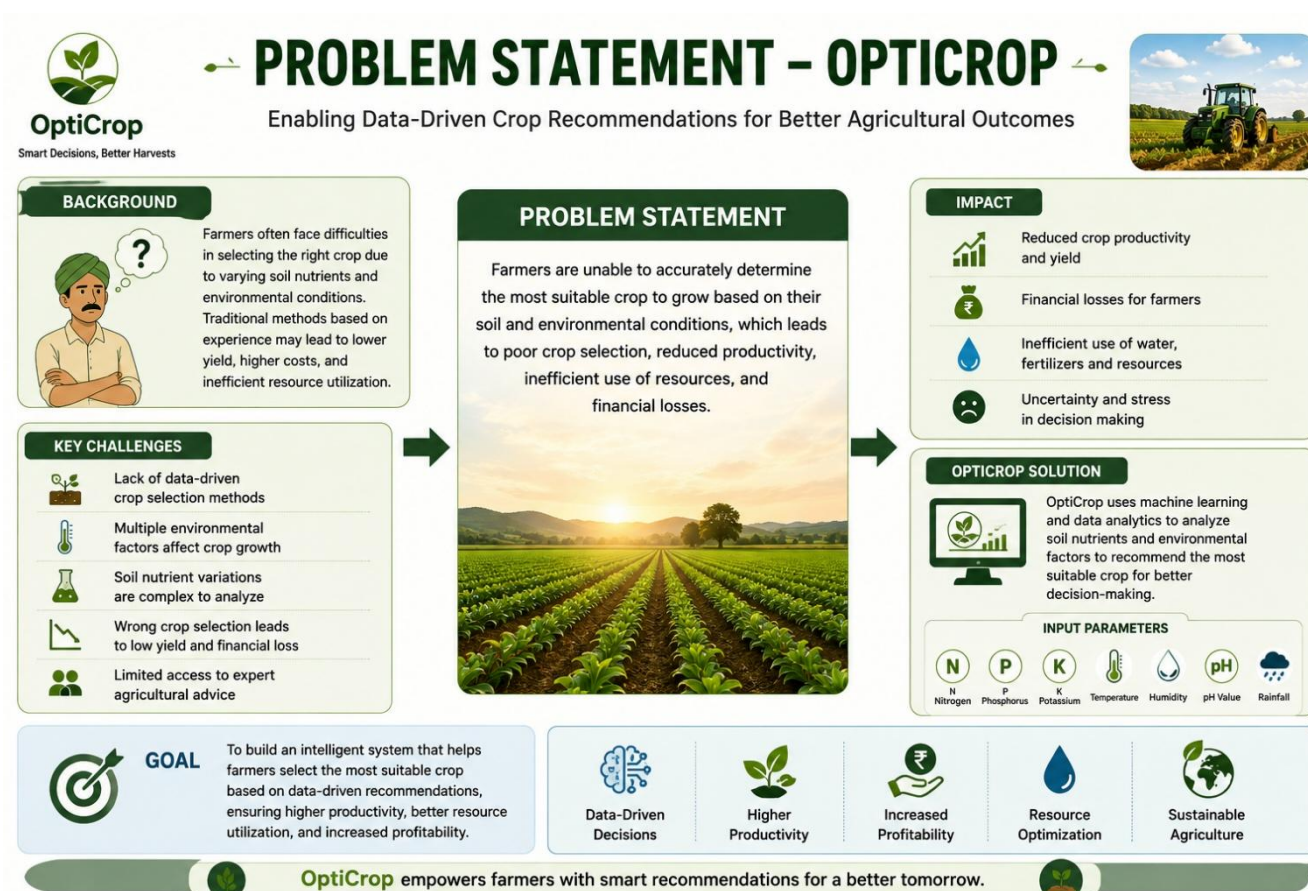
Project Initialization and Planning Phase

Date	29 June 2026
Team ID	xxxxxxx
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	3 Marks

Define Problem Statements (Customer Problem Statement Template):

Farmers often struggle to choose the right crop because of changing soil nutrients and weather conditions. Without proper guidance, they may select unsuitable crops, resulting in **reduced crop yield**, inefficient use of resources, and financial losses. OptiCrop uses machine learning to recommend the most suitable crop based on soil and environmental data, helping farmers improve productivity and make informed farming decisions.

Example:



GOAL

To build an intelligent system that helps farmers select the most suitable crop based on data-driven recommendations, ensuring higher productivity, better resource utilization, and increased profitability.

- Data-Driven Decisions
- Higher Productivity
- Increased Profitability
- Resource Optimization
- Sustainable Agriculture

Problem Statement (PS)	I am (Customer)	I'm trying to	But	Because	Which makes me feel
PS-1	A farmer	Select the most suitable crop for my farmland	I don't know which crop will provide the highest yield	Soil nutrients and weather conditions vary, and I lack expert guidance	Uncertain and worried about crop failure and financial loss
PS-2	A farmer or agricultural officer	Determine whether a specific crop is suitable for the current soil and climate conditions	I cannot accurately evaluate crop compatibility	Multiple environmental factors influence crop growth, making manual assessment difficult	Unsure about making the right cultivation decision
PS-3	An agricultural researcher or policymaker	Analyze crop suitability and agricultural trends	Processing large agricultural datasets is time-consuming and complex	Traditional methods lack predictive analytics and automation	Challenged to make timely, data-driven decisions for sustainable agriculture